



ELEC4142 – Transformer Protection

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Typical Arrangement for 132/11kV Transformer

Different Protection for Different Fault Types.

Main Protection

- 2 x Biased differential protection (BD)
- 1x HV restricted earth-fault protection (HVREF)
- 1 x LV restricted earth-fault protection (LVREF)
- High voltage winding instantaneous high set over-current (HSOC)
- Buchholz gas surge protection (Buchholz)
- Winding and oil temperature protection (WT/OT)

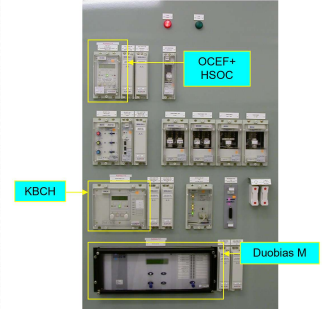
Backup protection

- Over current & earth fault protection (OCEF) at 132kV source
- For Parallel Transformer -
 - Directional OCEF protection (DOCEF)

Traditional transformer protection relays



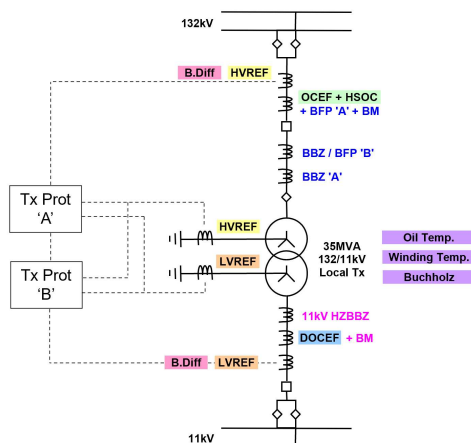
Integrated digital protection relays



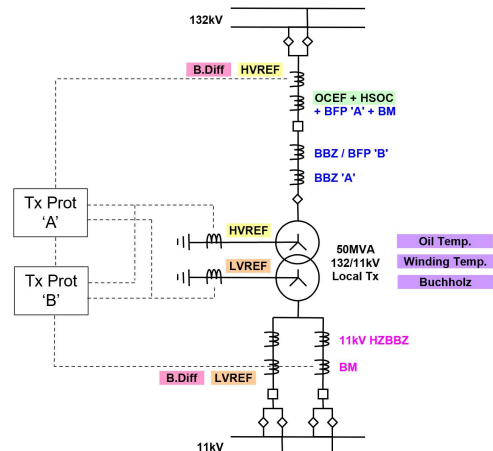
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Typical Arrangement for 132/11kV Transformer

35MVA Tx -



50MVA Tx -

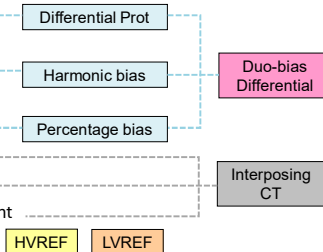


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Typical Arrangement for 132/11kV Transformer

Main protection

- Unit protection
- Able to cater for the following situations
 - Magnetizing inrush current
 - Current mismatch
 - Tap changer
 - CT ratio
 - Transformer ratio & phase shift
 - Flow of zero phase sequence current
 - Inadequate earth fault sensitive

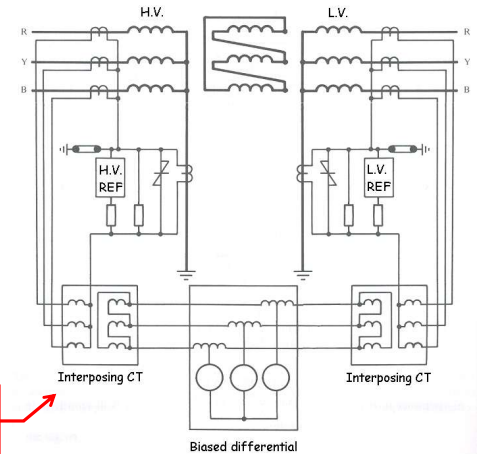


Backup protection

- OCEF + DOCEF

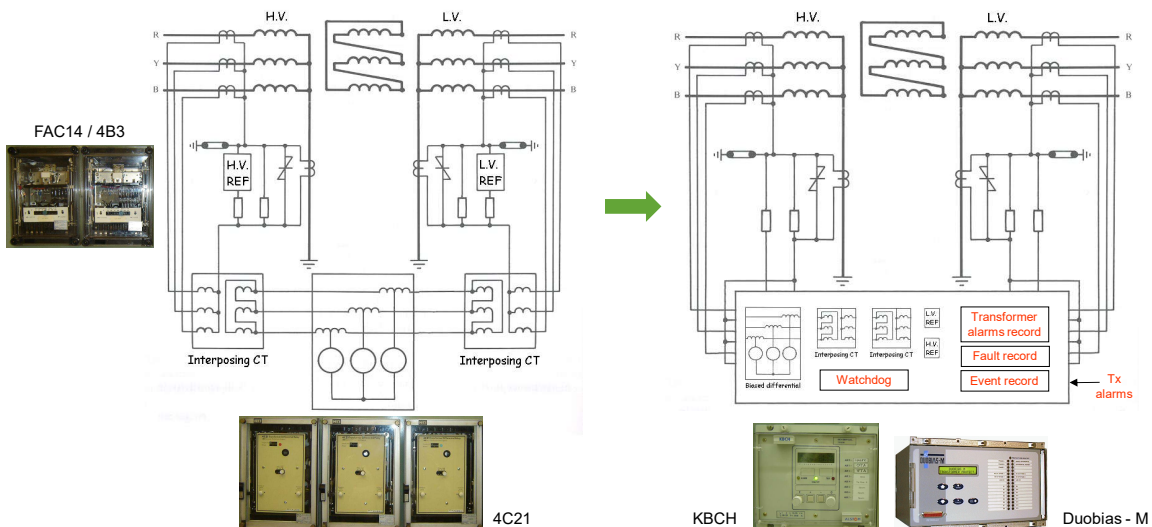
Additional protection

- Oil Temp / Wind Temp Trip + Buchholz Surge Trip (Tx / Tc)



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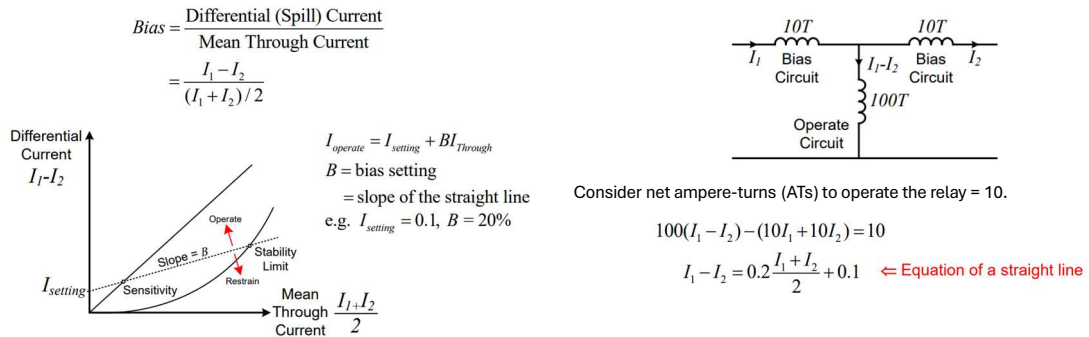
Typical Arrangement for 132/11kV Transformer – From E&M to Digital



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Bias Differential (BD)

- Consider a single-phase **biased differential** protection relay



- The increase in setting is therefore normally based on a **percentage of the through current**.
- Protection is also commonly referred to as **percentage biased differential**.
- Limit on allowable **spill** dictated by percentage of bias used
- Amount of spill is related to CT size**: stability limits and CT requirements are dependent on each other

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Bias Differential (BD) – Example

Question

- A 75 MVA 275/132 kV Yyn0 transformer is protected by Biased Differential. The tapping range of the on-load tap changer installed near the neutral on the 132kV side is $\pm 20\%$ referring to the nominal voltage. The relay setting available is 20%, 30% and 40%. The maximum through fault current is **1500A** at 275 kV side and is assumed constant over the whole tapping range (i.e. transformer impedance independent of tap changing).
- Determine a suitable current ratio for the interposing CT.
- Calculate a suitable bias setting for the relay in order to give a satisfactory coverage over the whole tapping range of the transformer under maximum through fault condition.

Answer

- Assume 200A is flowing in the 275kV side, current flowing in the HV side of CT.
- Secondary is therefore 1A. I_s is therefore $1/\sqrt{3} = 0.557A$ due to the delta connection of the interposing CT.
- Current flowing in the 132kV side is $200 \times 275/132 = 416.7A$. Current flowing in the secondary of the 400/1 CT is therefore **1.042A**.
- The ratio of Interposing CT is $1.042:1/\sqrt{3}$ or **1.042:0.577**. [Note: Interposing CT turn ratio]
- At +20% tap:

$V_{HV} = 275kV$ $V_{LV} = 132 \times 1.2 = 158.4kV$ $I_{HV} = 1500A$ $I_{LV} = 1500 \times 275/158.4 = 2604.2A$ $I_{2HV} = 1500/200 = 7.5A$ $I_{2LV} = 2604.2/(400 \times 1.042) = 6.5A$ $I_D = I_{2LV} - I_{2HV} = 1.25A$ $I_B = \min(I_{2HV}, I_{2LV}) = 6.5A$	At -20% tap: $V_{HV} = 275kV$ $V_{LV} = 132 \times 0.8 = 105.6kV$ $I_{HV} = 1500A$ $I_{LV} = 1500 \times 275/105.6 = 3906.25A$ $I_{2HV} = 1500/200 = 7.5A$ $I_{2LV} = 3906.25/(400 \times 1.042) = 9.77A$ $I_D = I_{2LV} - I_{2HV} = 1.875A$ $I_B = 7.5A$
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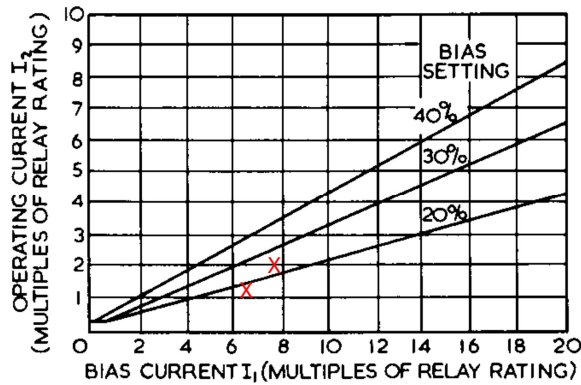
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Bias Differential (BD) – Example

Under Through Fault Condition -

- At +20% Tap: $I_D = |I_{2LV} - I_{2HV}| = 1.25 \text{ A}$ $I_B = 6.5 \text{ A} = \min(I_{2HV}, I_{2LV})$ $I_D / I_B = 0.192$
- At -20% Tap: $I_D = |I_{2LV} - I_{2HV}| = 1.875 \text{ A}$ $I_B = 7.5 \text{ A} = \min(I_{2HV}, I_{2LV})$ $I_D / I_B = 0.25$

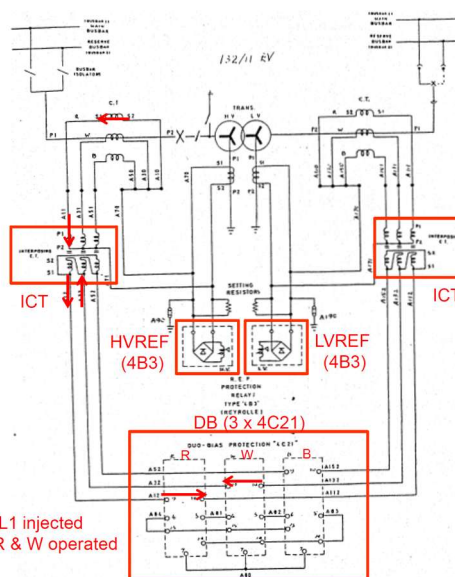
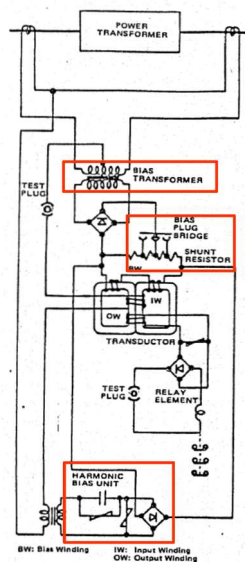
To maintain through fault stability, the bias differential relay should be set a **30% bias setting**.



Transformer Protection – E&M Relay (4C21)

Percentage Bias

Harmonic Bias
(Restraint)



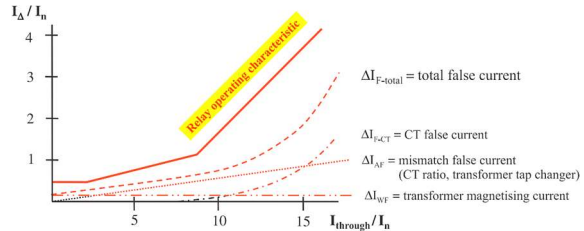
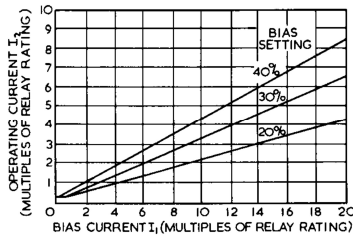
To replicate current
to unit value from
REF circuit.

HV L1 injected
→ R & W operated

132/11kV Transformer Protection – Percentage Bias

(1) **Percentage Bias** – maintain **stable for full load max/min tap with CT error and through fault condition** while remaining **sensitive for internal fault** with percentage differential plane (in multiple sections).

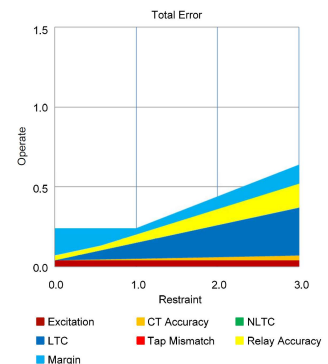
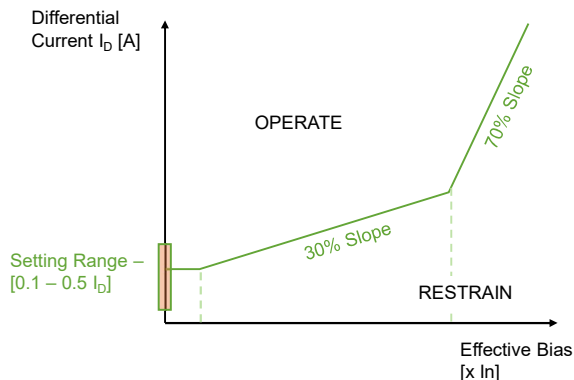
- **Steady State Differential** = Steady State Magnetizing Current
- **Percentage (a.k.a. Proportional) Differential** = Tap Mismatch + CT Error (under nominal load / through fault)
- **Transient Differential** = CT Saturation (due to high secondary voltage = symmetric saturation; due to DC offset = asymmetric)
- Line CT/ICT ratios and correction factors set to achieve current balance (1A into relay) at **nominal tap**
- Off nominal tap creates a differential, but it should be well below operating characteristics.
- Mal-operation due to this cause is avoided by selecting the **minimum bias** to be greater than sum of the maximum tap of the transformer and possible CT errors



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132/11kV Transformer Protection – Percentage Bias

Percentage Bias Curve -



- Bias in P643** to provide higher through fault stability:
- **Delayed Bias:** maximum bias quantity within last cycle.
 - **Transient Bias:** additional bias introduced with sudden increase in mean-bias, decayed exponentially.
 - **Maximum Bias:** maximum bias quantity in three phase[#]

First section is set higher than transformer magnetising current

(at steady state)

Second section is set to allow for off-nominal tap settings

Third section has a larger bias slope to cater for heavy through-fault conditions

[#] MAX {I_a, I_b, I_c} neglects any CT saturation in phase.

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132/11kV Transformer Protection – Percentage Bias

Manufacturer	Relay Type	Differential and Restraint Current		Remarks
Alstom	• P643	$I_D = \left \sum_i I_i \right $	$I_{RES} = \frac{1}{2} \sum_i I_i $	• Additionally included delayed bias , maximum bias and transient bias
Toshiba	• GRT100	$I_D = \left \sum_i I_i \right $	$I_{RES} = \frac{1}{2} \sum_i I_i $	
NR	• PCS978	$I_D = \left \sum_i I_i \right $	$I_{RES} = \frac{1}{2} \sum_i I_i $	• DPFC element uses $\Delta I_{RES} = \max \left\{ \sum_i \Delta I_{ai} , \sum_i \Delta I_{bi} , \sum_i \Delta I_{ci} \right\}$
SF	• CSC326	$I_D = \left \sum_i I_i \right $	$I_{RES} = \frac{1}{2} \left I_{j(MAX)} - \sum_{i \neq j} I_i \right $	• I_D is derived from fundamental frequency current, i.e. filtered quantities to avoid transient differential.
Siemens	• 7UT612	$I_D = \left \sum_i I_i \right $	$I_{RES} = \sum_i I_i $	• EFD (restraint blocking) block included until I_D stationary • DC component increases differential pickup with a factor of 2.
ABB	• RET512	$I_D = I_1 - I_2$	$I_{RES} = \max(I_1, I_2)$	• “ maximum restraint principles are less able to distinguish between internal and external fault with CT saturation as it ignores current contribution from all but one terminal zone.”

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132/11kV Transformer Protection – Percentage Bias

Setting Calculation

Given HWL Tx H1 information as follows:

50MVA 132/11kV Tx H1, HV CT = 1/400; LV CT = 1/1600; Tx Impedance = 27.0%, Tap Range = +5.25% to -19.25%; CT Error = 5%

1. ICT Ratio at Mid-Tap & Full Load Condition: (to ensure both side = 1A into relay)

$$I_{FL}[HV] \Big|_{\text{mid tap}} = \frac{50\text{MVA}}{\sqrt{3}(132\text{kV}) \left(1 + \frac{5.25\% - 19.25\%}{2}\right)} = 235.2\text{A}_{\text{pri}} \left(= \frac{235.2}{400} = 0.588\text{A}_{\text{sec}} \right) \rightarrow k_{CT1}[HV] = \frac{1}{0.588} = 1.701$$

$$I_{FL}[LV] \Big|_{\text{mid tap}} = \frac{50\text{MVA}}{\sqrt{3}(11\text{kV})} = 2624.3\text{A}_{\text{pri}} \left(= \frac{2624.3}{1600} = 1.640\text{A}_{\text{sec}} \right) \rightarrow k_{CT}[LV] = \frac{1}{1.640} = 0.6097$$

2. Differential Current after ICT at High, Low, Mid Tap with CT Error (-5%, +5%) (maintain stability under normal operating condition)

$$I_{FL}[HV] \Big|_{\text{lo tap}} = \frac{1}{400\sqrt{3}(132\text{kV})(1 - 19.25\%)} \times 1.701 = 1.152 \rightarrow (+5\%) = 1.210; (-5\%) = 1.094$$

$$I_{FL}[HV] \Big|_{\text{hi tap}} = \frac{1}{400\sqrt{3}(132\text{kV})(1 + 5.25\%)} \times 1.701 = 0.884 \rightarrow (+5\%) = 0.928; (-5\%) = 0.840$$

$$I_{FL}[LV] \Big|_{\text{any tap}} = 1.640 \times 0.6097 = 1.000 \rightarrow (+5\%) = 1.050; (-5\%) = 0.950$$

At Lo Tap with (+5% error at HV, -5% error at LV),

$$I_d = 1.210 - 0.950 = 0.260; \quad I_r = \frac{1.210 + 0.950}{2} = 1.080$$

It was found that the maximum differential with **CT error** would be at **Lo Tap** with (+5% error at HV, -5% error at LV), leading to $I_d = 0.260$; $I_r = 1.080$.

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132/11kV Transformer Protection – Percentage Bias

Setting Calculation (cont')

Criteria 1: To achieve stability under full load (FL) condition.

Slope of Small Current Region (p1):

$$p1 > \text{slope of max. diff current w.r.t. } (0,0) = \frac{I_{d FL \max}}{I_{r FL \max}} = \frac{0.260}{1.080} = 24.01\%$$

Select p1(setting) = 25.0% (must be greater than the calculated value)

Set Safety Margin of 50% to achieve stability under full load.

Let first section of operating curve as L1: $y = p1 \text{ } ir + ik$.

At $(y, ir) = (1.5 I_{d FL \max}, I_{r FL \max})$,

$$ik = y - p1 \times ir = 1.5 \times 0.260 - 0.25 \times 1.08 = 0.120$$

Select ik (setting) = 0.13 (must be greater than the calculated value)

Criteria 2: To achieve stability under loading effect during high impedance fault.

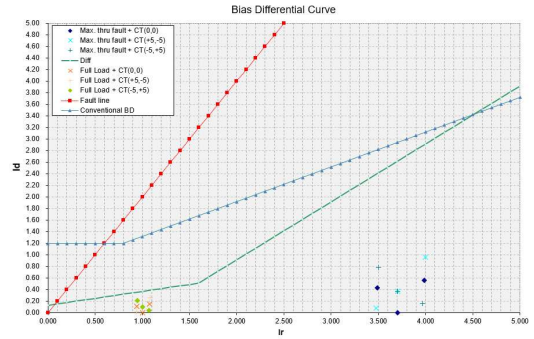
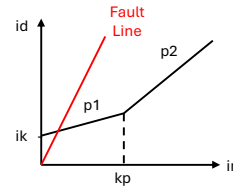
$$ir = 1.080$$

Define the turning point (id, ir) as (kp, ykp) such that $ykp = p1 \times kp + ik$

From Point Slope Form: Let $p2 = 100\%$ (smaller than the fault line slope = 200%)

$$kp = \frac{p2 \times Ir + ik}{p2 - p1} = \frac{100\% \times 1.08 + 0.13}{100\% - 25\%} = 1.613$$

Select Break Point with kp (setting) = 1.65.



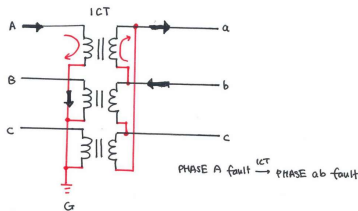
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132/11kV Transformer Protection – Interposing CT (ICT)

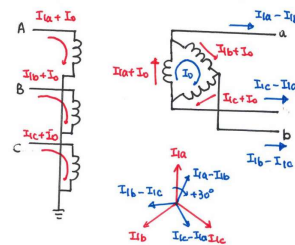
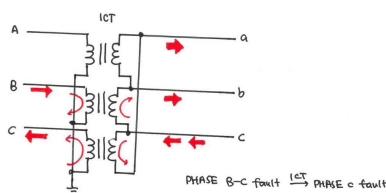
Interposing CT is intended to correct CT ratio, balance phase shift and filter out zero-sequence current.

In the case of a Y-delta transformer, an external fault will cause zero sequence current to be measured by the Y-side CTs. However, the zero-sequence current is trapped on the delta side, it is not measured by the delta-side CTs.

Earth Fault:



Phase Fault:



Zero Sequence Filtering

From WYE to DELTA with 30° clockwise:

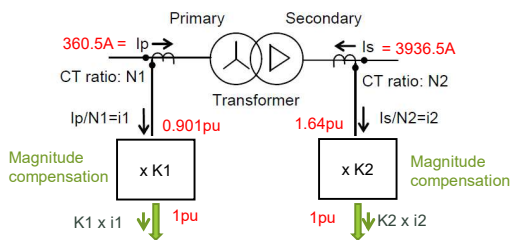
$$\begin{pmatrix} I_a' \\ I_b' \\ I_c' \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} I_a \\ I_b \\ I_c \end{pmatrix}$$

Zero Sequence Filtering:

$$\begin{pmatrix} I_a' \\ I_b' \\ I_c' \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} \begin{pmatrix} I_a \\ I_b \\ I_c \end{pmatrix}$$

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132/11kV Transformer Protection – ICT Corr Factor in Digital Relay



For 132/11kV 75MVA Tx with $Z_{TX} = 42.58\%$, tap range $+10.5 \sim -28.5\%$

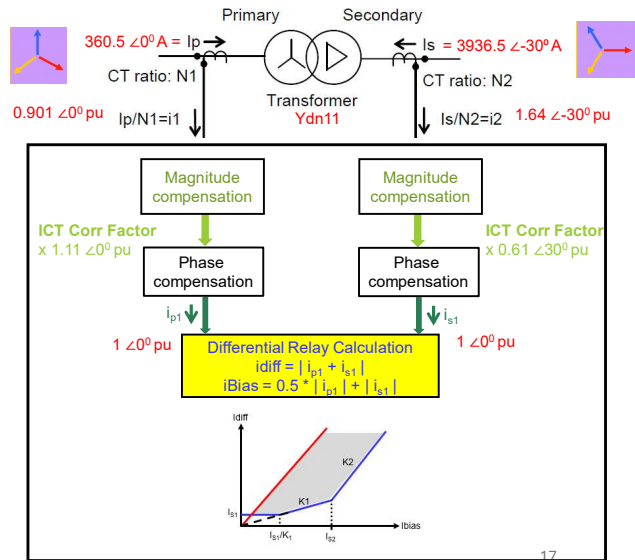
For Nominal Load Condition @Mid tap:

$$I_{FLHV} = \frac{75MVA}{\sqrt{3} \times 132kV(1 + \frac{10.5\% - 28.5\%}{2})} = 360.5A_{PRI} = 0.901A_{SEC}$$

Magnitude Compensation Factor $K_1 = 1A / 0.901A = 1.11$

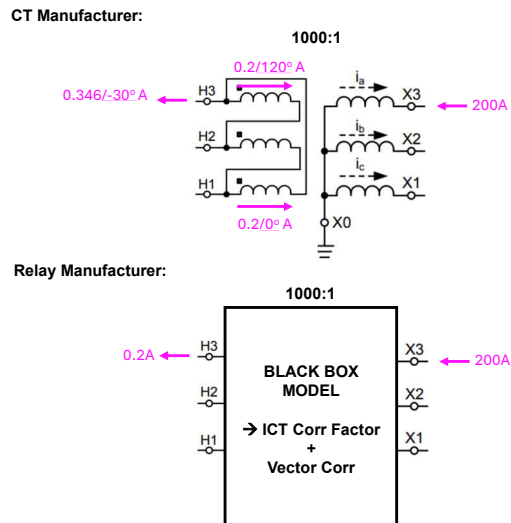
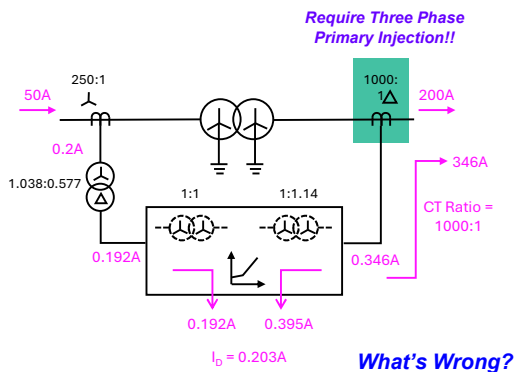
$$I_{FL_LV} = \frac{75MVA}{\sqrt{3} \times 11kV} = 3936A_{PRI} = 1.64A_{SEC}$$

Magnitude Compensation Factor $K_2 = 1A / 1.64A = 0.61$



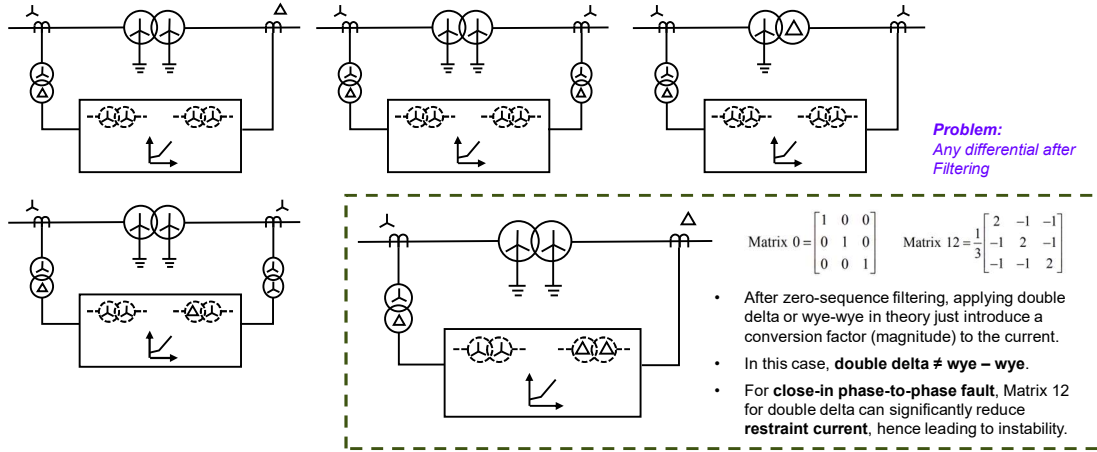
Incident 1 - Incorrect Setting Leading to Tripping

- Prior to the incident, the circuit were restored to service after 33kV cable circuit was diverted to the temporary switchboard. It was found with differential current under on load test, but the test result was confirmed with satisfactory. At 06:50, the circuit was tripped with incorrect setting.



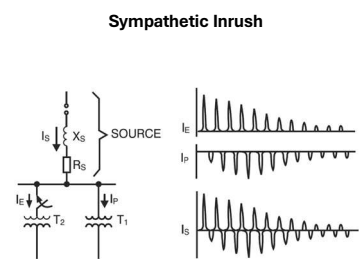
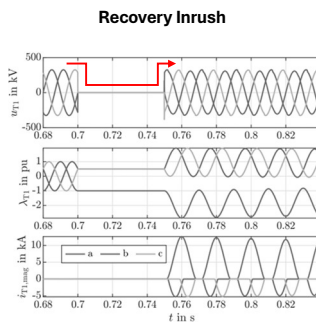
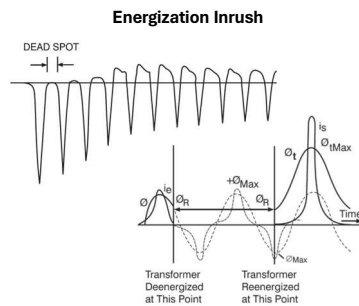
CT and ICT Connection Matching

- ICT aims to 1) convert the secondary output of CT to 1A at nominal load, 2) vector group matching at both ends, 3) zero sequence filtering and 4) duplicate current for another sets of protection.
- The following CT + ICT (+Tx) combination are equivalent.



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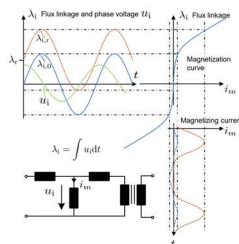
Transformer Inrush vs Fault



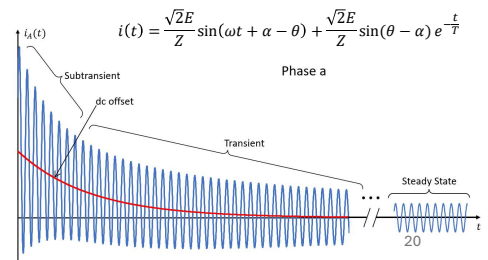
Inrush = Switched Event

$$\phi(t) = \begin{cases} L_m i(t) & 0 < t < t_s \\ L_s i(t) & t_s < t < \left(\frac{2(\pi - \theta)}{\omega} - t_s \right) \end{cases}$$

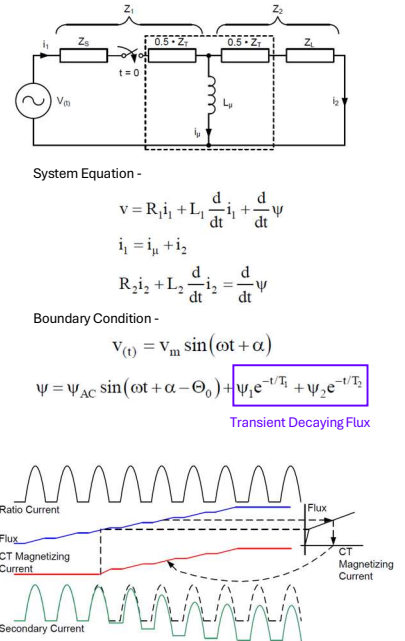
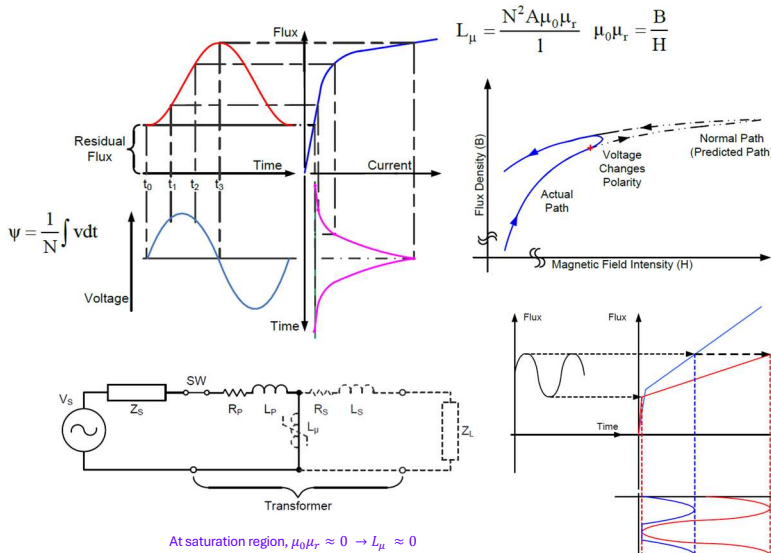
$$\phi(t) = \frac{V_m}{\omega} [-\cos(\omega t + \theta) + \cos(\theta)] + \phi_0$$



Fault



Transformer Physics -

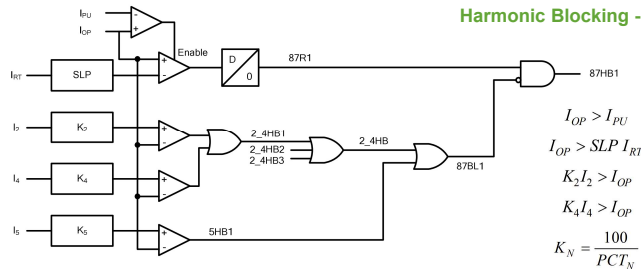


132/11kV Transformer Protection – Harmonic Bias

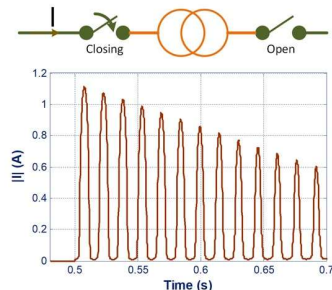
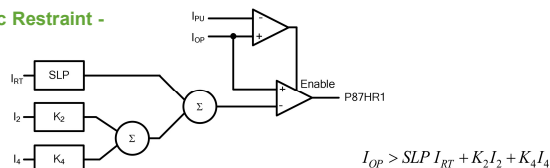
- (2) **Harmonic Bias** – prevent relay operation due to flow of pulses of **magnetizing inrush current** into one winding when the transformer is first energized

Note – 1) It is important to **detect fault upon energization**, so that the harmonic restraint function or blocking function should not stop fault detection and clearance.

- 2) 4C21 employs **cross-phase harmonic restraint**; while KBCH & DBM employs **harmonic blocking**.



Harmonic Restraint -

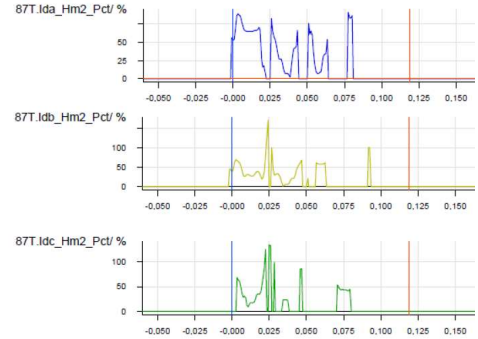
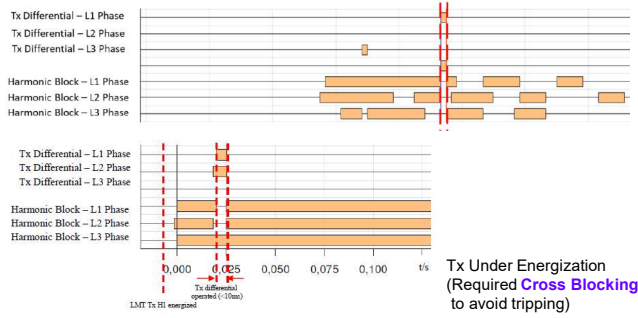


Incident 2: Transformer Tripped upon Energization with Harmonic Blocking

- Upon de-energization of a transformer through CB805, Tx differential relay 'A' operated, and Intertripping was initiated.
- Both Tx differential relays detected oscillation during transformer de-energization. second harmonic content of the oscillation was marginal relative to the harmonic restraint setting in use (<20%), and there was a short duration (<5 ms) where the **harmonic blocking elements** of the tripped relay reset for all 3 phases.
- Similar Case **under Energization** occurred on 2018.

Follow Up Action

- Set harmonic restraint setting of all bias differential relays to **15%**, especially for circuits with relatively long cables and circuits whose pre-commissioning tests were conducted before 2019.



External Fault Detection (EFD) with Transient Biasing

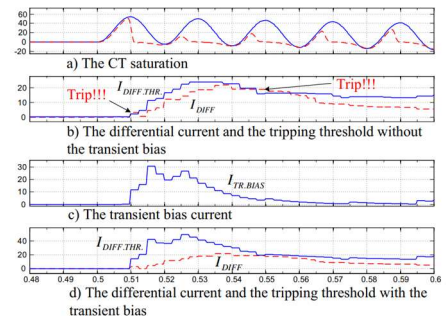
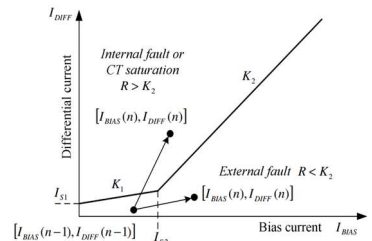
Transient bias is added to prevent false tripping of the protection system during external faults with CT saturation. When a **transient condition** is detected, the transient bias algorithm temporarily increases the tripping threshold of the differential protection.

1. This increase is based on the **positive ΔI_{BIAS}** . If the bias current increases, the tripping threshold is raised to prevent false trips.
2. The transient bias **decays quickly** once the transient condition subsides, ensuring that the protection system returns to normal sensitivity for detecting internal faults.

$$\begin{aligned} \Delta I_{DIFF}(n) &= I_{DIFF}(n) - I_{DIFF}(n-1), & \text{If } R < K_1 \text{ and } \Delta I_{BIAS}(n) > 0 \\ \Delta I_{BIAS}(n) &= I_{BIAS}(n) - I_{BIAS}(n-1), & I_{TR, BIAS}(n) = D \cdot I_{TR, BIAS}(n-1) + S \cdot \Delta I_{BIAS}(n), \\ R &= \Delta I_{DIFF}(n) / \Delta I_{BIAS}(n), & \text{otherwise} \\ & & I_{TR, BIAS}(n) = D \cdot I_{TR, BIAS}(n-1), \end{aligned}$$

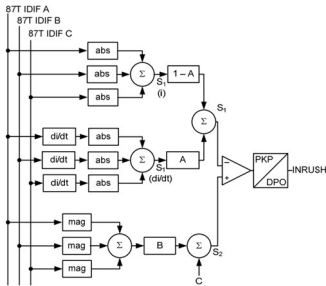
External Fault Detector (EFD) aims to prevent false trips during external fault and ensure quick and accurate detection of internal faults by maintaining the sensitivity of the differential protection.

- The system uses the ratio $R = \Delta I_{DIFF} / \Delta I_{BIAS}$ to detect external faults. If the ratio indicates an external fault, the transient bias is applied to stabilize the system.



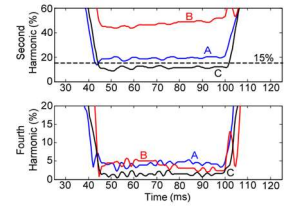
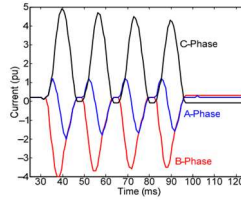
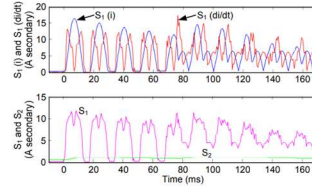
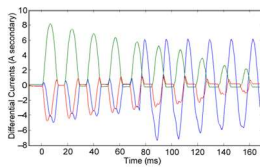
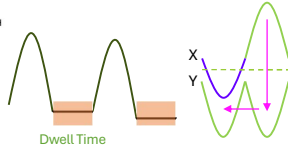
Inrush Detection

Dwell Time based Inrush Detection -

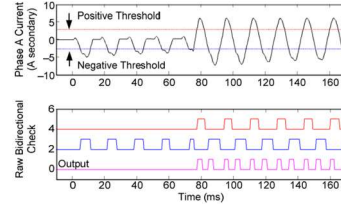
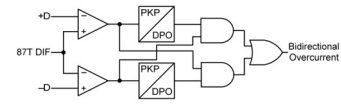


No Gap Detection: counts the numbers of consecutive samples below a dynamic threshold to determine the low current gap duration in the waveform (P643)

Wave Distortion Detection: determine the asymmetry with $K = |S_1| / |S_2|$, where $|S_1|$ is area of $X - Y$ and $|S_2|$ is area of $X + Y$. (PCS978)



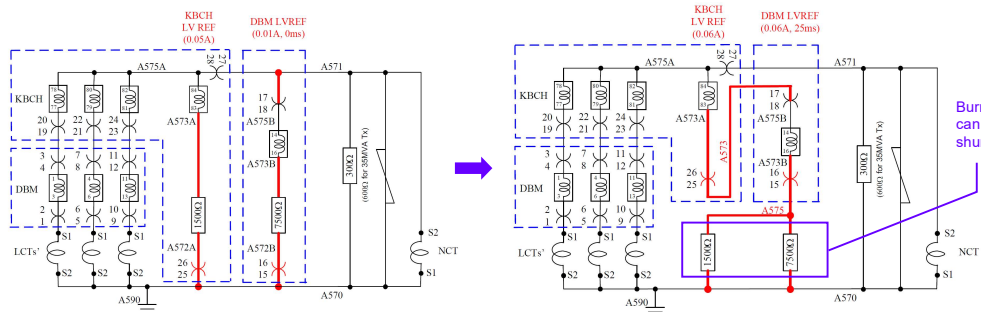
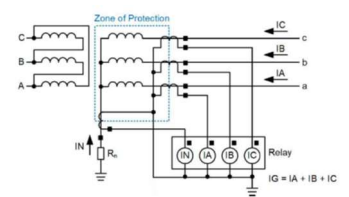
Bidirectional Overcurrent based Inrush Detection -



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132/11kV Transformer Protection – REF in High Impedance Principle

- After replacing the standard high impedance EM relay (4C21 for duo bias + FAC for REFs), the scheme with digital relays (KBCH + DBM) was in **high impedance principle**.
- Restricted Earth Fault (REF)** Protection is to enclose a “**restricted area**” with CTs such that the current sum should not be large under normal condition. However, with the **mismatch in line CT and neutral CT**, it is often that the neutral CT become saturated, hence leading to a differential current, and it is solved by high impedance principles.
- As illustrated below, another consideration is whether to put the REF element “**IN**” in **series** or **in parallel** after summing up the current. It was noted that even if the resistance was designed to obtain the **current ratio** to trigger the REF element of KBCH and DBM accordingly, the current flows between two parallel paths becomes not as expected with **resistor degradation**. Hence, the scheme was made series after that.



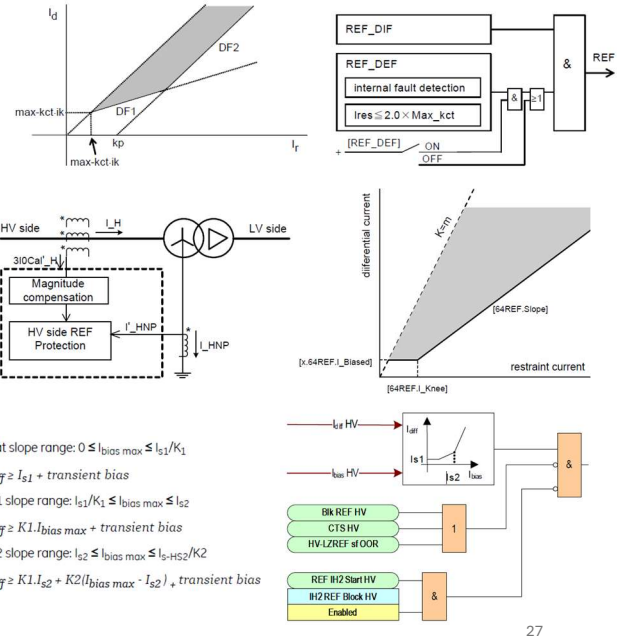
3

Low Impedance REF Element

Through Fault Stability of REF element is prone to CT saturation. Different manufacturers require different **blocking** or **restraining** mechanism on such component.

Relay	Differential and Restraint
GRT100	$I_D = K_1 I_{01} + I_N$ $I_R = \max(K_1 I_A , K_1 I_B , K_1 I_C , K_0 I_N)$
PCS978	$I_{0D} = K_1 I_{01} - K_N I_N$ $I_{0R} = \max \{ I_{01}, I_N \}$ * CT Saturation Blocking with 2 nd (for transient) and 3 rd harmonics (for steady state) and CT transient blocking $I_0 < \beta_0 \times I_1$ enabled.
P643 [#]	$K = N\text{-CT Ratio} / \text{ph-CT Ratio}$ $I_D = (I_A + I_B + I_C) + K I_N = 3I_0 + K I_N$ $I_R = 0.5 (\max(I_A , I_B , I_C) + K I_N)$ * Transient Bias and Delay Bias enabled.
7UT613	$I_D = 3I_0 $ $I_R = k (3I_0 - I_N - 3I_0 + I_N)$ * Depending on Phase Relationship

[#] additional restraint + delay timer at operation
= "restraint blocking"



132/11kV Transformer Protection – REF in Low Impedance Principle

Setting Calculation

Given a 35MVA 132/11kV Transformer ($Z_{TX} = 27.0\%$, $+10.5\%$ to -24.0%) with maximum through fault = 129.63MVA (746.03A at Tap Low of 132kV) and full load = 35MVA (201.43A at Tap Low of 132kV). The HV CT ratio is 400/1.

The differential current caused by CT error is independent of Tap Position.

Full Load Current at Tap Low:

$$I_{FL} = \frac{35\text{MVA}}{\sqrt{3} \times 132\text{kV} (1 - 24\%)} = 201.43\text{A} (= 0.504\text{A}_{sec})$$

Max. Differential under Full Load Condition (due to CT error):

$$I_{d,FL} = 10\% \times I_{FL} = 10\% \times 0.504\text{A} = 0.05$$

$$I_{r,FL} = 105\% \times I_{FL} = 105\% \times 0.504\text{A} = 0.529$$

Criteria 1: Full Load Stability

1p1 (Slope 1) = 10% [Fixed] Sensitivity = 20% of Tx Rated Current

$1\text{ik} = I_{FL} \times 20\% = 0.504 \times 20\% = 0.101 \rightarrow$ Select **1ik = 0.11**

Criteria 2: Through Fault Stability

Fault Line Slope = 100% $\rightarrow$ Select **1p2 (Slope 2) = 50%** [Fixed]

$1\text{kp} > \text{max full load current} = 0.504 \rightarrow$ Select **1kp = 0.51**

Checking – Stability under Full Load and Max Through Fault Condition:

	I_d [A]	I_r [A]	$\log$ [A] at I_r	% of $\log/I_d$
Full load	0.050	0.529	0.143	283.72%
Max. thru fault	0.187	1.958	0.724	388.28%

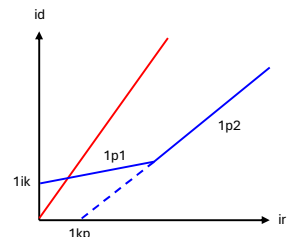
Maximum Through Fault at Tap Low:

$$I_{MTF} = \frac{35\text{MVA}/27\%}{\sqrt{3} \times 132\text{kV} (1 - 24\%)} = 746.03\text{A} (= 1.865\text{A}_{sec})$$

Max. Differential under Max. Through Fault Condition (due to CT error):

$$I_{d,MTF} = 10\% \times I_{MTF} = 10\% \times 1.865\text{A} = 0.187$$

$$I_{r,MTF} = 105\% \times I_{MTF} = 105\% \times 1.865\text{A} = 1.958$$

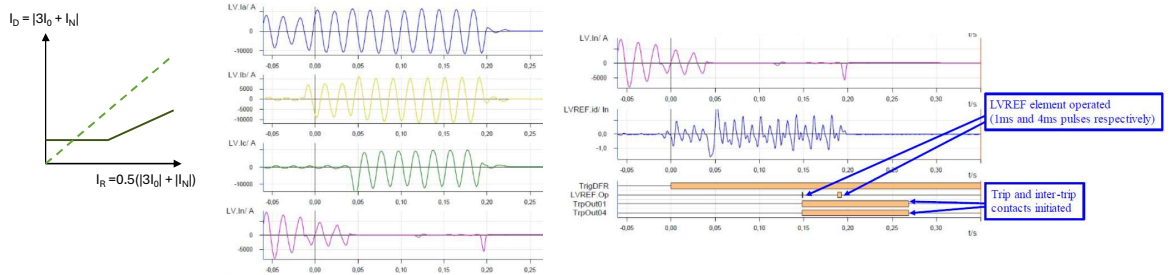


Incident 3: LVREF Unstable Leading to Through Fault Tripping

- In 2023, a transformer tripped simultaneously under **through fault** with **LVREF element**.
- From the fault records in one transformer differential relay, current measurement (both 3-phase and neutral) of the relay were normal and found matched with those from the another differential relay and fault recorder. However, it was observed that the **LVREF element (low impedance, percentage differential type)** of the tripped relay operated twice at very **short pulses** of 1ms and 4ms respectively.
- It is concluded that the mal-operation of the tripped relay should be caused by the relay internal logic on LVREF detection upon high level of **through fault current**.

Follow Up Action

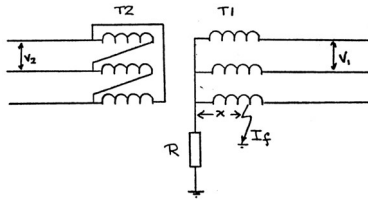
- Change the **delay timer setting** for LVREF to 25ms to ride-through any inrush.
- Apply blocking function ("**Harmonic Blocking of REF**" and "**Positive Sequence Current Block**").



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Additional Consideration on Winding Coverage

- Consider a resistive earthed delta-wye transformer.



If the earthing resistor is rated to pass through full load current I_{FL} ,

$$\frac{V_1}{\sqrt{3}R} = I_{FL}$$

Assume $V_1 = V_2$, then $T_2 = \sqrt{3}T_1$.

For a fault at x p.u. distance from the neutral,

$$I_F = x \frac{V_1}{\sqrt{3}R}$$

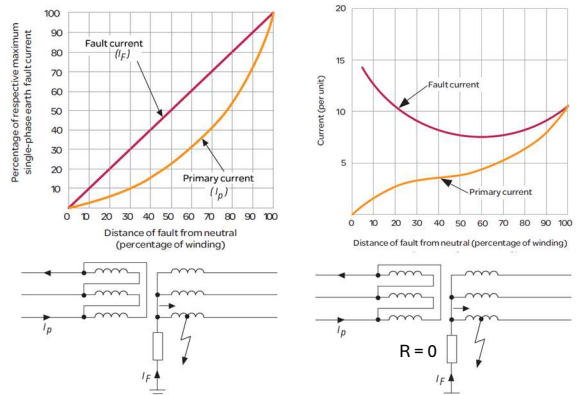
Effective turn ratio = T_2/xT_1

$$I_{Primary} = x \frac{V_1}{\sqrt{3}R} \frac{xT_1}{T_2} = \frac{x^2}{\sqrt{3}} I_{FL}$$

$$CT \text{ secondary current (on primary side of Tx)} = \frac{x^2}{\sqrt{3}}$$

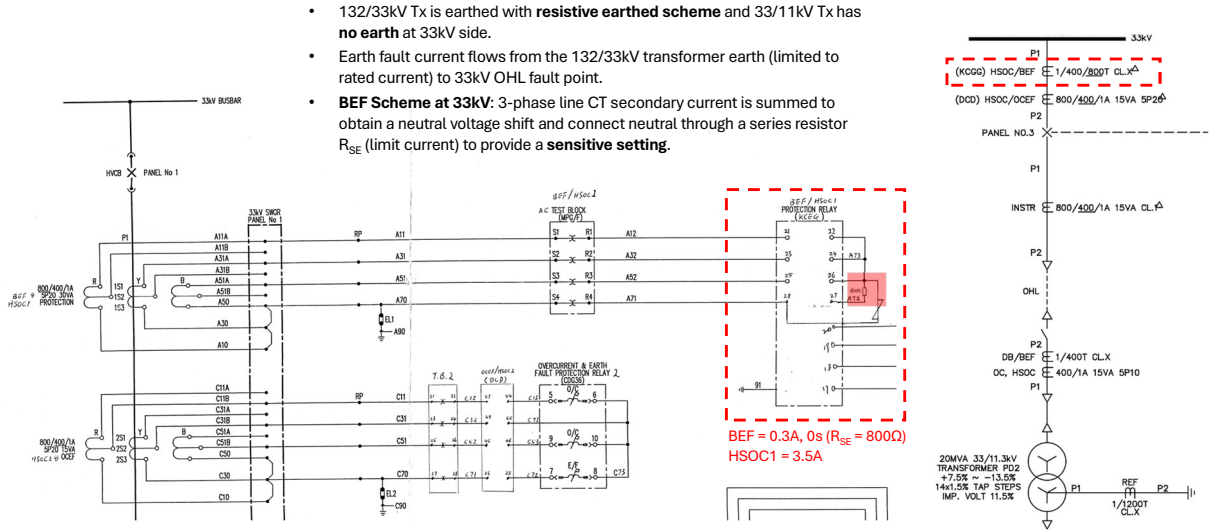
If differential setting = 20%, then $\frac{x^2}{\sqrt{3}} > 20\% \rightarrow x > 59\%$

i.e. 59% of winding is unprotected.



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Balanced Earth Fault (BEF)



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132/11kV Transformer Protection – HSOC Setting

SETTING Criteria:

- At a suitable margin above worst 11kV through fault to ensure stability.
- At a level high enough to ensure stability for worst case inrush current during transformer energization
- At a level sensitive enough to ensure fault detection for the min. fault level condition on 132kV side
 I_F = Maximum through fault current

Example

- Maximum Through Fault (11kV side) = 185.2MVA
- Maximum Through Fault Current (11kV side) = $810A \times \frac{185.2MVA}{\sqrt{3} (132kV)}$
- 132kV Maximum Fault Level = 6000MVA

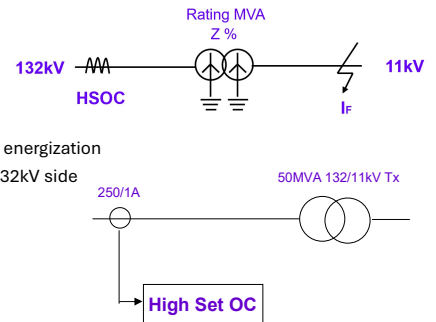
- CT Secondary Current = $810/250 = 3.24A$
- HSOC safety factor is 2.2

- HSOC setting = $3.24 \times 2.2 = 7.12A$

In general, safety factor = 1.3 ~ 1.5 for MHJ, CAG39, KCGG & DCD
= 2 ~ 2.2 for CAG37

(To prevent mal-operation under relay transient overreaching)

$$\text{Safety Factor} \times I_{F, \text{MAX}} (11kV) < \text{HSOC Setting} < I_{F, \text{MIN}} (132kV)$$



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Incident 4: Transformer HSOC Tripped on Through Fault

- A transformer was tripped on through fault ($I_F = 766A$) by HSOC (Setting = 788A) in 2005.
- It was found that the **maximum through fault at maximum tap** = 680A from system parameter calculation. Hence, the HSOC setting 788A should be acceptable. The existing transformer impedance parameter was in doubt.
- Upon site transformer impedance test conducted in 2005, it was found that the transformer impedance values provided from transformer test report was given on base of **tap voltage & current**, instead of **nominal voltage and current** as expected.

Tap Position	Injected HV Current [A]	Injected HV Voltage [V]	Measured Impedance at 30°C [%]	Calculated Impedance with Ref. to Nominal Tap 8 [%]	Factory Test Impedance at 75°C [%]
1	1.334	395.2	28.13	34.36	28.06
8	1.754	395.6	26.16	26.16	26.03
13	2.152	395.7	24.92	21.33	N/A
24	3.507	395.4	22.63	13.08	22.48

Note –

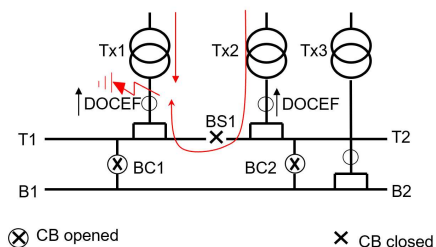
The calculated Impedance is based on nominal Tap 8 (132kV & 153.1A) as the calculation based for all tappings.

Recall Base Change Formula:

$$Z_2 = Z_1 \left(\frac{V_{1B}}{V_{2B}} \right)^2 = 22.63 \left(\frac{1 - 24\%}{1} \right)^2 = 13.08 \quad \text{Lowest Tap} = -24\%$$

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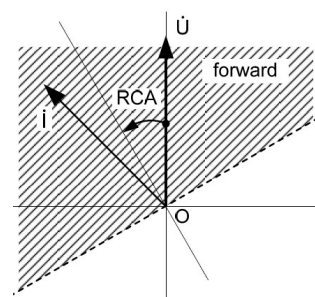
132/11kV Transformer Protection – DOCEF



- Transformers** can be operated in parallel in case **fault level** allows.
- DOCEF provides **backup** for **uncleared upstream fault** in case **inter-tripping** does not work.
- Operate with “**forward**” current, i.e. current going back to 132kV.
- Direction is given by the comparison between voltage and current phasor
- During **VT fuse burnt condition**, the voltage phasor becomes **unreliable**.
- Setting: OC = 0.3A (very sensitive)

Phase of Protection	Operate Current (I)	Polarizing Voltage (U)
A Phase	Ia	Ubc
B Phase	Ib	Uca
C Phase	Ic	Uab

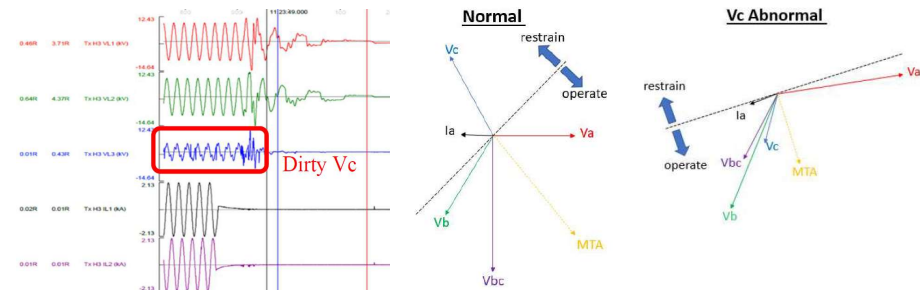
- Directional forward
 $-90^\circ < (\text{angle}(I) - \text{angle}(U) - \text{RCA}) < 90^\circ$
- Directional reverse
 $-90^\circ > (\text{angle}(I) - \text{angle}(U) - \text{RCA}) > 90^\circ$



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Incident 5: DOCEF Operative with Fuse Burnt

- Prior to the incident, **HV VT fuse** for a transformer was blown out. Planned outage was scheduled for **VT fuse replacement** at 01:00. At 19:24, the transformer **tripped on DOCEF protection** and all other ends of the RMU circuit were inter-tripped.
- From fault recorder waveform retrieved, **dirty voltage signal** with significant deformation in both **magnitude** and **phase angle** could be observed in L3 voltage. Owing to the abnormality in L3 voltage, the forward region of directional element would be shifted accordingly, resulting in the operate current marginally within the forward operation region.
- In case of VT primary fuse deterioration -
 - 1) Request SOD to shut down the tx for VT fuse replacement, or
 - 2) Request SOD to re-configure busbar config so that the Tx with VT problem could run in parallel with other Tx (to cater for DOCEF undesired operation), or
 - 3) Issue a TLR to state "possible unstable of DOCEF protection", or
 - 4) Defeat the concern DOCEF protection if the tx is running as independent auto (IA) mode, consider it is a backup protection of the tx main protection.



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132/11kV Transformer Protection – Buchholz

Gas Production causes oil to be displaced and causes the float to drop and operate the switch for alarm purpose. Another switch is calibrated to operate at a certain oil flow rate (surge) to trip the transformer CB.

Buchholz Gas

- Gas generation may be a **result of** the following:
 - Decomposition / degrading of solid or liquid insulation inside the transformer due to the **overheating** or **arcing**.
 - From the **outside** towards the pipeline.
 - From the **oil itself** due to unsatisfactory de-gassing prior filling.
- If **sufficient gas** is found to **displace** the upper float, the **alarm (BGA)** should be **activated**.
- The **small valve** at the top left is to bleed the gas off and **reset** the relay.

Buchholz Surge

- Rapid oil movement in the pipeline **towards the conservator** is caused by an **internal arc short circuit**, or **hot spot** which must be correctly addressed.
- Oil leaks from the transformer are **environmentally unacceptable** and a **fire hazard** will lead to transformer failure.

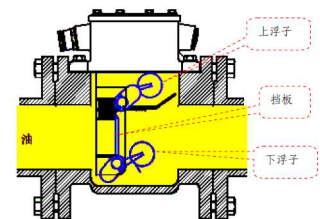


图1-2-1-14 正常情况下，气体继电器状态示意图

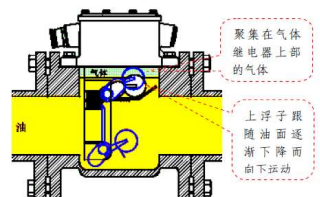


图1-2-1-15 气体继电器内聚集气体示意图

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132/11kV Transformer Protection – Buchholz

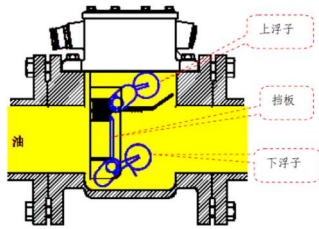


图1-2-1-14 正常情况下，气体继电器状态示意图

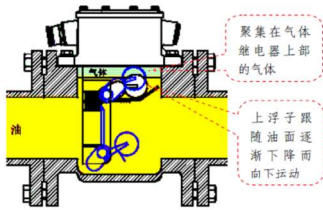


图1-2-1-15 气体继电器内聚集气体示意图

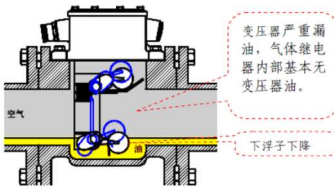


图1-2-1-16 气体继电器内部无油示意图

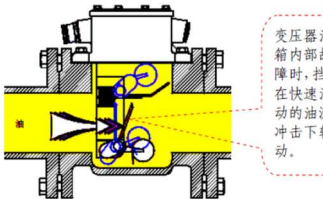
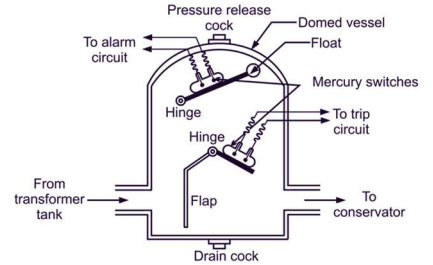
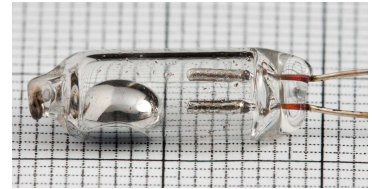
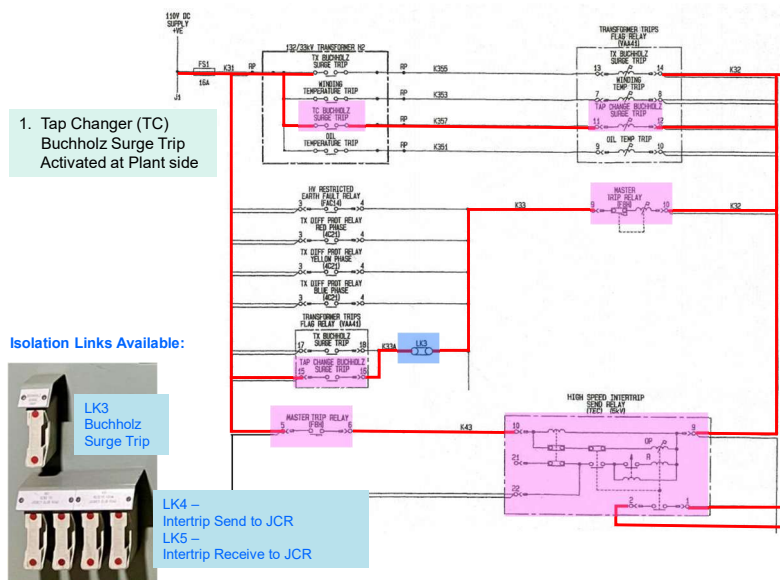


图1-2-1-17 油流冲击气体继电器挡板示意图

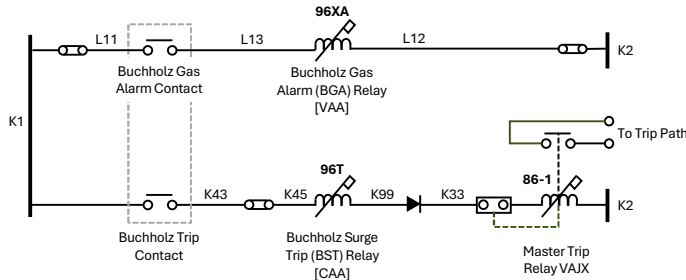


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Buchholz Surge Trip (BST) Arrangement – Voltage Operate Repeat

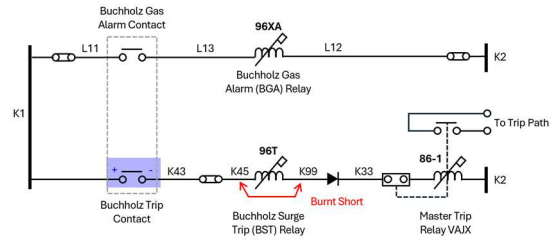
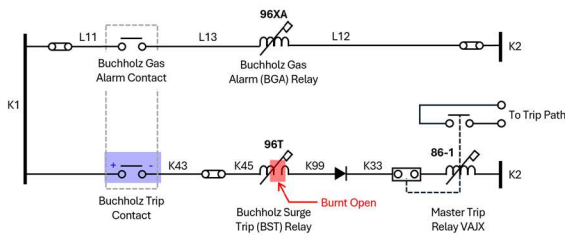


Buchholz Surge Trip (BST) Arrangement – Current Operate Repeat



At 400/132kV Tx, the BST element triggers a CAA relay (for alarm purpose only) through its coil in series with the trip relay.

It allows **short time tripping** (to fulfill requirement on critical clearance time) and it ensures **stability under open circuit / short circuit** of BST relay.



132/11kV Transformer Protection – Tripping Allocation

Tx Fault Trip Relay / Tx Master Trip Relay (For local Tx)

Initiated by:

- HSOC
- HV REF
- LV REF
- Transformer Biased Differential Protection
- Main Tank & Tap changer Buchholz surge Protection

Actions taken:

- Trip LV CB & HV CB
- 11kV Auto-Switching Initialization (ASI)
- For RMU fed Tx
 - If HV CB is present → Trip HV CB
 - If HV CB is absent → High Speed Inter-trip (HSI)



Tx LV Trip Relay

Initiated by

- OCEF
- Oil Temperature Trip
- Winding Temperature Trip

Actions Taken:

- Trip LV CB
- For RMU fed Tx
 - If HV CB is present → Trip HV CB + HV CB BFP
 - If HV CB is absent → Remote Trip Initiation (RTI)

Auto-Switching is initiated by OTT, WTT separately

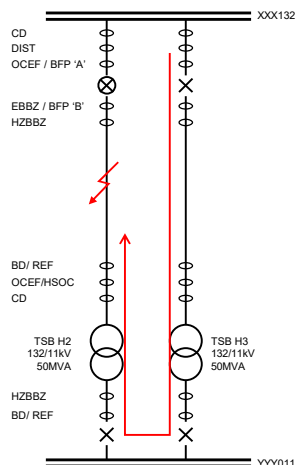


Remote Trip Initiation (RTI)

- Initiated by
 - 11kV Busbar Protection
 - Transformer LV Trip
- Operation
 - Current check (same CT for OCEF/HSOC) + Timer (0.3s)
 - Auxiliary contact (Tx-RMU LBS)
- Action Taken
 - Intertrip whole 132kV RMU ring



Back Energization – What is the problem in view of Protection?



Requirement on Protection Philosophy EP-0048 may not be fulfilled.

1. 132kV requires duplicated main protection.

Cable Fault at Circuit with 132kV open: only CD operative to trip LV CB
DIST inoperative with CT position.

Transformer Fault at Circuit with 132kV open: only BD operative to trip LV CB
HSOC inoperative with limited fault current.

Note: Fault level after going through 2 transformers in series is around 95MVA (and possibly smaller).

OCEF Setting: OC = 75%, 0.6 (Operation Time at Fault Level = 12.9s)
EF = 30%, 0.5 (Operation Time at Fault Level = 2.8s)

CD Requires OC Guard = 1.0A and EF Guard = 0.3A.

OCEF activates 11kV CB trip directly and HSI Send.

CD activates also Master Trip relay to initiate 11kV CB trip, HSI Send and ASI.

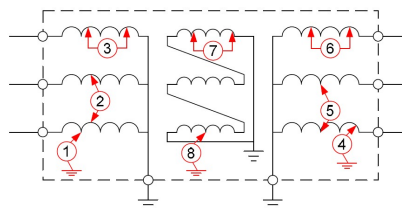
2. 132kV requires duplicated DC supplies.

Note: Normal operation with DIST source at 132kV + CD source at 11kV constitutes duplicated DC supplies.

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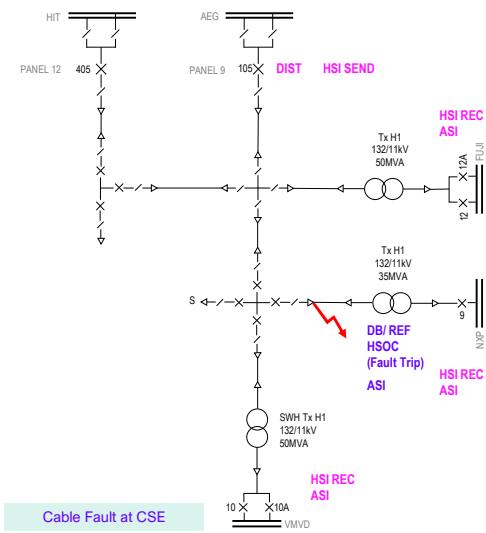
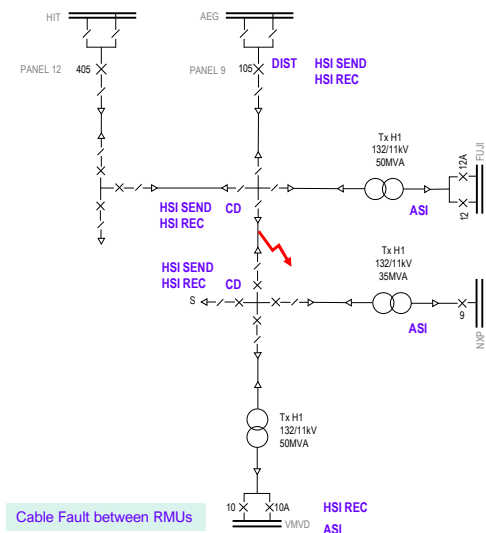
132/11kV Transformer Protection – Protection Summary

Fault Type	Protection Used
Pri. winding ph-ph fault	Differential, OC+HSOC
Pri. winding ph-E fault	Differential, HVREF, EF
Sec. winding ph-ph fault	Differential
Sec. winding ph-E fault	Differential, LVREF
Interturn/Core/Tank fault	Buchholz
Overheating	Oil & winding temp.
Reverse current	DOCEF (35MVA Only)



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Actual Protection Operations under Different Faults



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Secondary Injection – Precommissioning Test in KBCH + DBM

KBCH Bias Differential Injection

□ Differential setting test for initial setting = 20%:

Phase	Cal. op. current	* Operating current	LED	LCD	Relay contact check RL3 (42-44)	Relay contact check RL7 (41-43)
HV A-B	A	A	□ Trip	F n- I G 1 A * B * C * F AUX01234567Io123	□	□
HV B-C		A	□ Trip	F n- I G 1 A * B * C * F AUX01234567Io123	□	□
HV C-A		A	□ Trip	F n- I G 1 A * B * C * F AUX01234567Io123	□	□
LV A-B	A	A	□ Trip	F n- I G 1 A * B * C * F AUX01234567Io123	□	□
LV B-C		A	□ Trip	F n- I G 1 A * B * C * F AUX01234567Io123	□	□
LV C-A		A	□ Trip	F n- I G 1 A * B * C * F AUX01234567Io123	□	□

* Pick-up current error = +/- 5%. Drop-off current $\geq 0.9 \times$ Pick-up current.

* For 20MVA Tx (for ph-to-ph injection):

HV op. current (secondary) = Diff. initial setting / 2 / 1.155 = (0.20/9) / 2 / 1.155 = 0.096A

LV op. current (secondary) = Diff. initial setting / 2 / 0.618 = (0.20/9) / 2 / 0.618 = 0.180A

* For 50MVA Tx (for ph-to-ph injection):

HV op. current (secondary) = Diff. initial setting / 2 / 0.981 = (0.20/9) / 2 / 0.981 = 0.113A

LV op. current (secondary) = Diff. initial setting / 2 / 0.352 = (0.20/9) / 2 / 0.352 = 0.316A

For 35MVA Tx (for ph-to-ph injection):

HV op. current (secondary) = Diff. initial setting / 2 / 0.866 = (0.20/9) / 2 / 0.866 = 0.128A

LV op. current (secondary) = Diff. initial setting / 2 / 0.387 = (0.20/9) / 2 / 0.387 = 0.287A

DBM Bias Differential Injection

* Pick-up current error = +/- 5%. Drop-off current $\geq 0.9 \times$ Pick-up current.

* For 20MVA Tx (for ph-to-ph injection):

HV op. current (secondary) = Diff. initial setting / 2 / 1.155 = 0.35 / 2 / 1.155 = 0.152A

LV op. current (secondary) = Diff. initial setting / 2 / 0.618 = 0.35 / 2 / 0.618 = 0.283A

* For 50MVA Tx (for ph-to-ph injection):

HV op. current (secondary) = Diff. initial setting / 2 / 0.866 = 0.4 / 2 / 0.866 = 0.231A

LV op. current (secondary) = Diff. initial setting / 2 / 0.312 = 0.4 / 2 / 0.312 = 0.641A

For 35MVA Tx (for ph-to-ph injection):

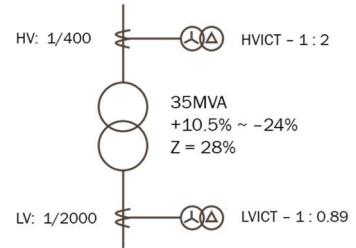
HV op. current (secondary) = Diff. initial setting / 2 / 0.866 = 0.5 / 2 / 0.866 = 0.289A

LV op. current (secondary) = Diff. initial setting / 2 / 0.387 = 0.5 / 2 / 0.387 = 0.646A

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Secondary Injection – Precommissioning Test in KBCH + DBM

Consider a 132/11kV 35MVA transformer with the following information
Find the **phase-to-phase injection current** for KBCH relay to operate.



Solution:

At mid – tap

HV: 132kV $(1 + (10.5\% - 24\%)/2) = 123\text{kV}$

LV: 11kV

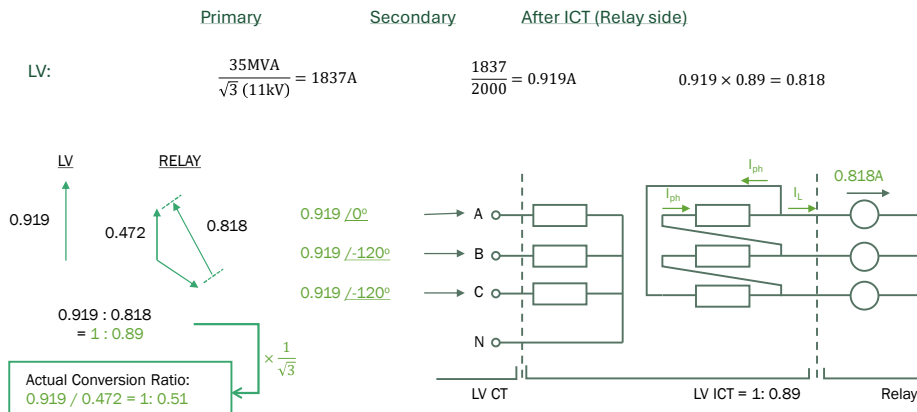
	Primary	Secondary	After ICT (Relay side)
HV:	$\frac{35\text{MVA}}{\sqrt{3} (123\text{kV})} = 164.17\text{A}$	$\frac{164.17}{400} = 0.410\text{A}$	$0.410 \times 2 = 0.820$
LV:	$\frac{35\text{MVA}}{\sqrt{3} (11\text{kV})} = 1837\text{A}$	$\frac{1837}{2000} = 0.919\text{A}$	$0.919 \times 0.89 = 0.818$

$\text{LV CT: } 1/2000$ $\text{LV ICT: } 1:0.89$

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Secondary Injection – Precommissioning Test in KBCH + DBM

What does it mean by **LV ICT = 1: 0.89** in KBCH relay?

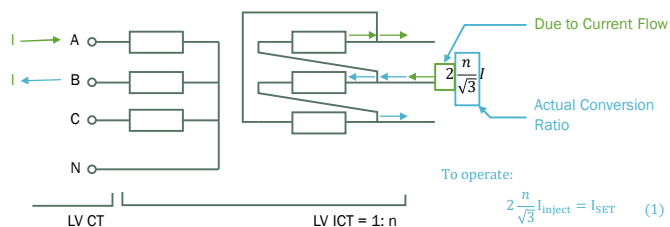


LV ICT at KBCH means **input line current : relay line current = 1A : 0.89A** at balanced load.

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Secondary Injection – Precommissioning Test in KBCH + DBM

Phase to Phase Injection



As there is a fixed $I_{SET} = [\text{DIFF. INIT.}]$

- Hence,
For HV: $n_{HV} = 2$,

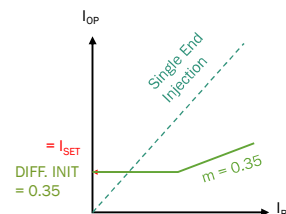
$$I_{\text{inject}} = \frac{I_{\text{SET}}}{2 \frac{n}{\sqrt{3}}}$$

For LV: $n_{LV} = 0.89$,

$$I_{\text{inject,HV}} = \frac{0.35}{2 \frac{\sqrt{3}}{3}} = 152\text{mA}$$

For LV: $n_{LV} = 0.89$,

$$I_{\text{inject,LV}} = \frac{0.2/0.9}{2 \frac{0.89}{\sqrt{3}}} = 216\text{mA}$$

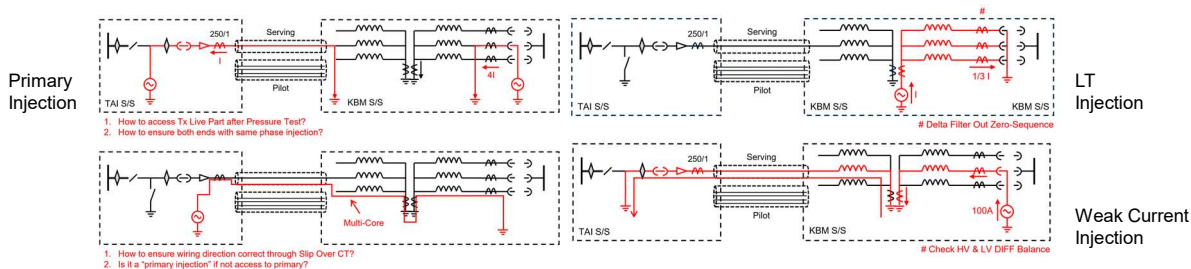


Commissioning Test for Transformer Protection

- After **secondary injection** to prove protection integrity (wiring, relay logic, function), a **primary injection test** is performed to ensure primary current reflected to secondary through CT (**ratio, polarity, connection, balance**) as expected.
- Primary Injection** is referred to injection through primary (possibly with wires through slip over CT) to test HV and LV CT balance; while **LT Injection** (Low Tension Injection) is referred to injection to test line and neutral CT balance, possibly proving N CT integrity after CT replacement.

Problem

- Lack of Three Phase Injection Set with High Burden
- Transformer in High Impedance with Large CT Ratio (H1: 53Ω; HV CT = 250/1, LV CT = 1000/1)
- Different Locations for CTs (Injection through conductor or {pilot / cable sheath})
- Access to Primary Conductor after Pressure Test (Earth Bar Disconnection)



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Transformer Commissioning – Energizing 3 Times

Example -

First Energization

- Set BC OC protection timer to 0.1 (clear **M1BB** for energization)
- Grant ROT to M1 and R1 BB (in case the HZBBZ and EBBZ in the panel is not proven)
- Energize in stage to RMU, then to transformer (first energization) at T/C = 1
- Close 11kV CB to energize M1BB and Close CB to energize Mobile Cap Bank.
- Check VT Output normal, phase rotation correct; CVT Phase Comparison Test between Tx and Mobile Cap Bank
- 132kV HZBBZ and EBBZ stability test in turn; Feeder Protection on load stability; RMU CC on load stability
- De-energize. Perform on load transfer to clear **R1BB** for energization

Second Energization

- Put T/C to IN at Tap 8 and Energize in stage to transformer. Put T/C to IA. Close CB to energize Mobile Cap Bank
- Transformer on load stability test;
- 132kV HZBBZ and EBBZ stability test in turn; Connect transient recorder to perform on load transfer test in turn for HZBBZ and EBBZ.
- 11kV HZBBZ on load stability test for Tx and BC panel

Third Energization

- Phasing out with another transformer

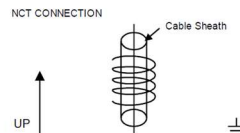
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Additional Tests under Pre-Commissioning Test

8. Transformer Neutral Cable Sheath Insulation Test

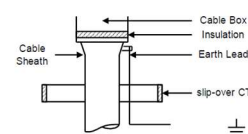
- ☐ Test results copied from record in PCS978 test form.
- ☐ 2kV megger test for 1 minute = _____ MΩ (>2 MΩ)
- ☐ If megger test result is below 2 MΩ, carry out a.c. pressure test at 2kV r.m.s for 1 minute
- ☐ Tx. neutral cable sheath is earthed through the detachable earth link.

To avoid sheath current cancelling neutral conductor current leading to incorrect results.



9. Slip-Over Line CT Cable Sheath Check

- ☐ Test results copied from record in PCS978 test form.
- ☐ Check cable sheath earthing arrangement for all slip-over line CTs according to PGG 2/00-045.

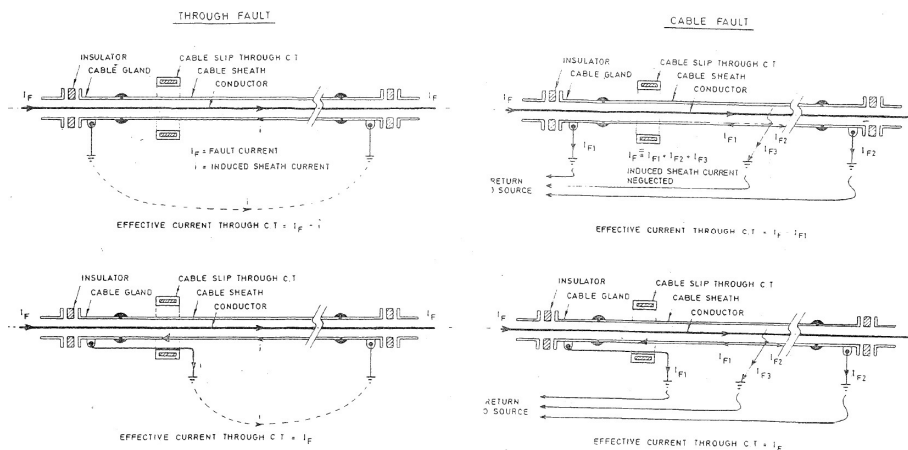
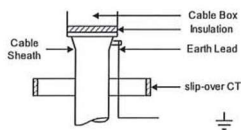


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Earthing at Cable Sheath

- Under fault conditions, this **induced voltage** can become dangerously high. Therefore, it's a standard safety requirement to **earth the cable sheath or armour at both ends**. Earthing can provide a **complete path for induced current** to flow, which may lead to incorrect measurement of fault current for CTs if the lead is incorrectly applied.

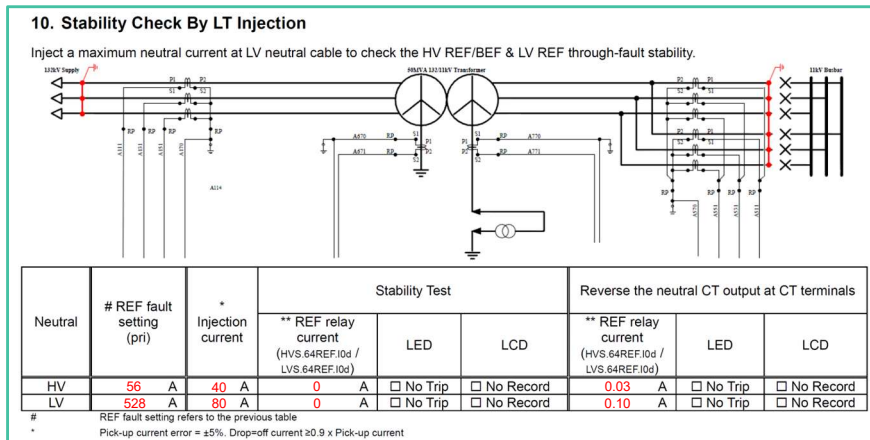
Pre-Commissioning Test Form: Slip-over Line CT Cable Sheath Check



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LT Injection

- LT Injection** is performed under **SFT** to prove the **CT integrity** (neutral CT polarity and spill) and **connection** (e.g. tap changer, winding, cable end box) after **CT replacement** (neutral / phase) or **transformer replacement** with **injection at primary**.
- The **actual connection** is subject to the **burden** (transformer impedance), **accessibility of test point** (e.g. transformer neutral-to-earth connection, transformer phase (if exposed), gap between CT and neutral cables / phase conductor)
- It was performed as a pre-commissioning test after **function test** and **secondary injection**.

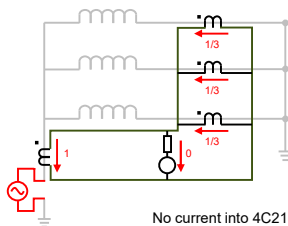


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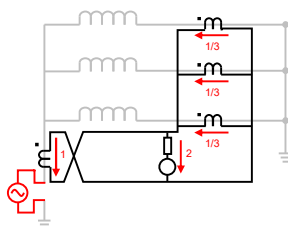
LT Injection

REF Setting	
35MVA Tx	50MVA Tx
HV: 400/1 POC = 30A	HV: 400/1 POC = 56A
LV: 2000/1 POC = 300A	LV: 1600/1 POC = 530A

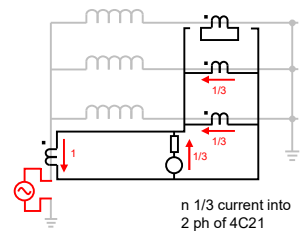
Normal Stability



Crossed Neutral Condition



Short-and-Isolate Phase Condition

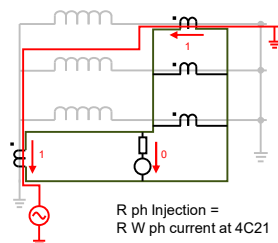


Method 1: Injection at Neutral

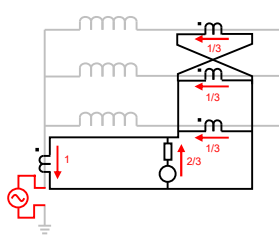
- Primary Connection (CT & Tx Connection) Proven
- May not have large enough current to be seen at secondary

Method 2: Injection through M/C

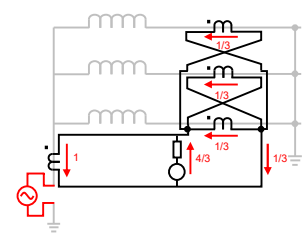
- Large enough current to go through (~ 100A at primary)
- Require Accessible CT for M/C to go through
- Primary Connection not Proven (rely on correct M/C connection)



Issue: 1 CT Crossed Condition



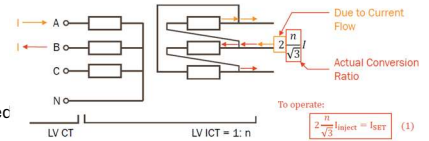
Issue: 2 CT Crossed Condition



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On Load Stability Test

- To put in service a transformer protection with Commissioning Test with instruction issued from SCE as record.



Bias Differential Element

		□ 35MVA Tx (Note 1)		□ 35MVA Tx (1A HV/LV 5A) (Note 2)		□ 50 MVA Tx (Note 3)		□ 50MVA Xian Tx (Note 4)		□ 75MVA Xian Tx (Note 5)		□ 50MVA ABB Tx (Note 6)		□ 120MVA Tx (Note 7)	
		HV op. current	LV op. current	HV op. current	LV op. current	HV op. current	LV op. current	HV op. current	LV op. current	HV op. current	LV op. current	HV op. current	LV op. current	HV op. current	LV op. current
Ph-Ph	Secondary Injection	0.071A	0.159A	0.071A	0.159A	0.09A	0.256A	0.076A	0.213A	0.187A	0.341A	0.102A	0.284A	0.100A	0.094A
	Primary Injection	28.5A	317.8A	28.5A	63.56A	36A	409A	30.6A	340.7A	74.9A	817.8A	40.8A	454A	60.0A	70.5A
Ph-E	Secondary Injection	0.143A	0.318A	0.143A	0.318A	0.18A	0.511A	0.153A	0.426A	0.374A	0.681A	0.204A	0.568A	0.199A	0.188A
	Primary Injection	57A	635.6A	57A	127.1A	72A	817.8A	61.1A	681.5A	149.8A	1635.5A	81.5A	909A	199.4A	141A

Note 1 For 35MVA Tx (for ph-to-ph injection)

HV op. current (secondary) = Diff. Initial setting * $\sqrt{3} / 2.43 / 2 = 0.20 * 0.356 = 0.071A$
 HV op. current (primary) = [HV op. current (secondary)] * CT ratio = $0.071 * 400 = 28.5A$
 LV op. current (secondary) = Diff. Initial setting * $\sqrt{3} / 1.09 / 2 = 0.20 * 0.795 = 0.159A$
 LV op. current (primary) = [LV op. current (secondary)] * CT ratio = $0.159 * 2000 = 317.8A$

Note 4 For 50MVA Tx (for 1-ph-to-E injection)

HV op. current (secondary) = Diff. Initial setting * $\sqrt{3} / 2.43 = 0.20 * 0.713 = 0.143A$
 HV op. current (primary) = [HV op. current (secondary)] * CT ratio = $0.143 * 400 = 57A$
 LV op. current (secondary) = Diff. Initial setting * $\sqrt{3} / 1.09 = 0.20 * 1.589 = 0.318A$
 LV op. current (primary) = [LV op. current (secondary)] * CT ratio = $0.318 * 2000 = 635.6A$

$1/\sqrt{3}$ due to star-to-delta
 2.43 due to ICT ratio
 /2 due to current flow (phase-to-phase)

Restricted Earth Fault Element

		□ 35MVA Tx		□ 35MVA Tx (1A HV/LV 5A)		□ 50 MVA Tx (Include Xian & ABB)		□ 75MVA Xian Tx		□ 120MVA Tx	
		HVREF op. current	LVREF op. current	HVREF op. current	LVREF op. current	HVREF op. current	LVREF op. current	HVREF op. current	LVREF op. current	HVREF op. current	LVREF op. current
Ph-E	Secondary Injection	0.10A	0.18A	0.10A	0.9A	0.14A	0.33A	0.23A	0.33A	0.20A	0.17A
	Primary Injection	40A	360A	40A	360A	56A	528A	92A	792A	120A	127.5A

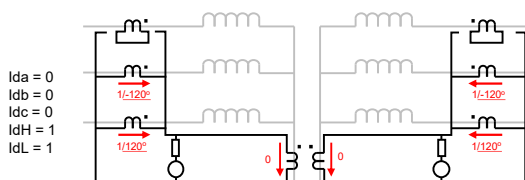
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On Load Stability Test

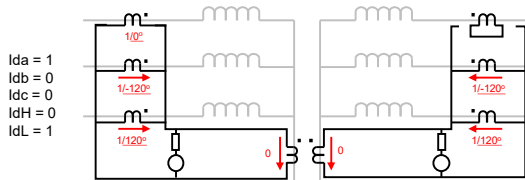
Note:

- Protection Isolation should be well executed before test to avoid tripping
- The following test steps are for digital relay which requires to test BD and REF element simultaneously. For EM relay, one should consider the condition similar to LT injection.

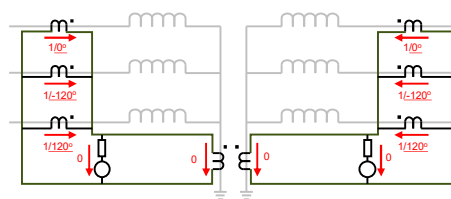
REF Condition 1



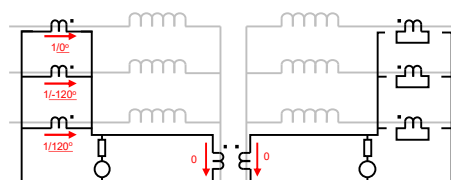
REF Condition 2



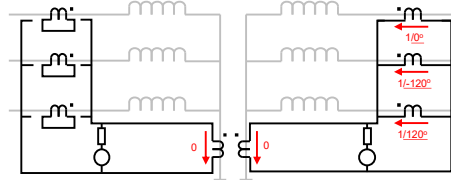
Normal Condition



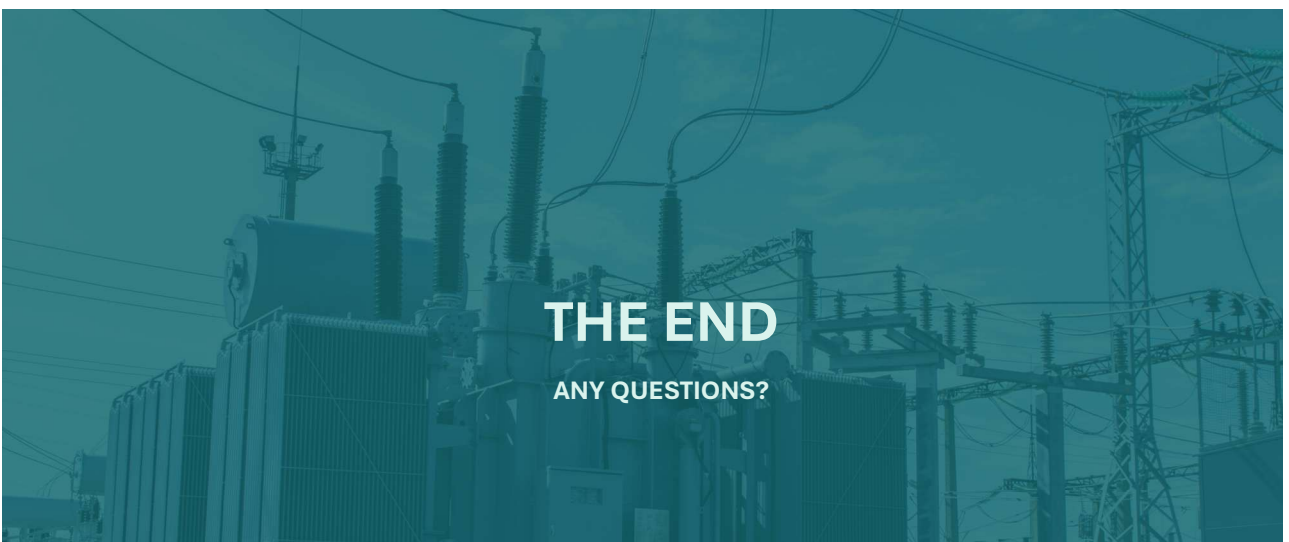
BD Condition 1



BD Condition 2



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THE END

ANY QUESTIONS?

Power System Analysis and Protection Studies

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